Keerti Vijay Ananth

Mobile | Email | LinkedIn | Portfolio | GitHub

Experience

Motorola Solutions *Software Engineer Intern* Feb 2026 – Oct 2026

- Contributed to the system design of a fault-tolerant, multi-data-center messaging architecture for **mission-critical communications**, using an **N+1 redundancy model** so the remaining data centers absorb the full load if one site fails, with no service interruption.
- Evaluated **RabbitMQ, Temporal, and NATS** for reliable message delivery, workflow orchestration, and horizontal scaling, focusing on preserving critical messages during failover and traffic growth.
- Built a Proof of Concept and designed **CI/CD pipeline** workflows that simulated data center failures, validating failover behavior and enabling **zero-downtime deployments**, with documented findings to guide architectural decisions.

BVK Exports *AIML Data Analyst Intern* May 2025 – July 2025

- Built Python-based data pipelines using Pandas and NumPy to clean and structure multi-source datasets, reducing processing time by **40%**.
- Applied machine learning techniques (clustering, segmentation) to uncover customer patterns, improving targeting efficiency by **30%**.
- Optimized SQL queries and automated workflows to create scalable data systems, delivering actionable insights through dashboards.

Projects

NULLDECK – Touchless AI DJ Workstation | Live Demo Oct 2026

- Built a browser-based **two-deck DJ workstation** that replaces a physical controller with real-time webcam hand tracking, letting users crossfade, set per-deck volume, and trigger cue points through hand position and pinch gestures.
- Developed the system using **React, TypeScript, and the Web Audio API** for live two-channel audio mixing, with a **FastAPI and librosa** backend for BPM, key, and section detection, and **MediaPipe Hands** for gesture recognition.
- Integrated two independently licensed, fully mixable audio sources (local upload and the **Jamendo Creative Commons** catalog) alongside **Spotify OAuth** for playlist discovery, keeping playback within each platform's audio-usage terms.

TrialLens – Clinical Trial Intelligence Tool | Live Demo Aug 2026

- Engineered a **5-step LangGraph workflow** routing natural-language queries across SQL lookups, keyword-based document retrieval, and FDA adverse-event APIs, using **Llama 3.3 70B (Groq)** for intent parsing and cited response synthesis.
- Integrated **ClinicalTrials.gov and openFDA FAERS APIs**, implementing full-text search fallbacks for condition matching and resolving production API access issues caused by TLS fingerprinting and WAF restrictions.
- Optimized production deployment by replacing a **Chroma + sentence-transformers** retriever with a lightweight TF-IDF-style scorer, eliminating the ML runtime and resolving **512MB OOM failures**, while deploying the Dockerized service on Render with provider-agnostic LLM support.

Advanced Diabetes Prediction System Sep 2025

- Engineered a supervised ML pipeline using Scikit-Learn to classify diabetes risk levels from clinical and lifestyle attributes with an accuracy of **94.5%**.
- Developed modular Python scripts for feature engineering, model training, hyperparameter tuning, and performance validation.
- Integrated real-world inference through a React + Node.js interface backed by AWS SageMaker, enabling instant risk-level predictions and actionable health insights.

Education

Vellore Institute of Technology Bhopal, India

B.Tech (CSE) Health Informatics - 8.65 CGPA *Sep 2022 – Oct 2026*

Co-curricular

- Presented the paper "Analysis of Nanomaterial-Based Biosensor Health Monitoring Systems with Artificial Intelligence for Smart Healthcare" at ICINVENTS-25, SKCET.

Technical Skills

Languages: Python, Java, Go
AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision (YOLO), NLP, LLMs,RAG, LLM Fine-tuning
Data Science & Analytics : Feature Engineering, Pandas, NumPy, Statistical Analysis, Scikit-Learn
Message Brokers: Temporal, NATS, RabbitMQ
Developer Tools: Power Bi, Tableau, Linux, Git, Github, MySQL, Docker

Achievements

PR Lead & Outreach – Health-O-Tech Club (Techno club @ VIT Bhopal) 2023 - 2025
Winner - Ecom-Thon @ ADVITYA Techno festival Feb 2022
Assisted in making of Python Tamil E-Content Course at VIT Bhopal Nov 2023